

Banning a Vanishing Margin: Brazil’s Child Marriage Ban, Formal Marriage, and Informal Unions*

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Abstract

In 2019 Brazil eliminated all legal pathways to civil marriage below age 16. We evaluate the reform using a prespecified age-eligibility difference-in-differences design and three sources that measure distinct margins: civil-marriage registrations, coresident unions, and mothers’ conjugal status at birth. The preferred registry specification estimates a 1.1% decline at age 15 (95% CI $[-14.9\%, +14.9\%]$); omitting age-specific trends yields -37.8% . PNADC estimates a suggestive 0.40 percentage-point relative increase in coresident-union prevalence (95% CI $[-0.003, +0.801]$), but cannot establish informal substitution. SINASC shows that formal marriage among mothers under 16 had flattened before 2019; its trended and no-trend estimates span -23.9% to $+29.3\%$ and fail placebo diagnostics. The evidence neither identifies an additional short-run fall in formal marriage nor establishes no effect. It shows why administrative declines, union formation, and legal formalization must be evaluated separately.

JEL: J12, J13, K36, O15, I38. **Keywords:** child marriage; minimum-age laws; informal unions; civil registration; Brazil.

*Working draft; please do not cite or circulate. Replication package: all estimates derive from a specification frozen before estimation (SHA-256-locked on 2026-09-02, before any post-reform coefficient was computed) and reproducible end-to-end with a single command. [ACKNOWLEDGMENTS TO BE ADDED]

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1 Introduction

Minimum-age-of-marriage laws are the standard policy instrument against child marriage. More than fifty countries have tightened them since 2000, and ending child marriage by 2030 is an explicit target of the Sustainable Development Goals. The evidence on whether these laws change behavior, however, is mixed at best: enforcement is weak in most of the developing world [Collin and Talbot, 2023], Mexico’s state-level bans reduced *registered* child marriages while informal unions took their place [Bellés-Obrero and Lombardi, 2023], and an information experiment in Bangladesh found that publicizing a child-marriage reform could even backfire [Amirapu et al., 2026]. The core difficulty is that the law regulates a legal act—civil marriage—while the phenomenon of concern is a living arrangement that in much of the world requires no state involvement at all.

Brazil’s Law 13,811 of March 2019 offers an unusually sharp version of this problem. The law closed the last legal pathways to civil marriage below age 16, which until then had been available in cases of pregnancy or to avoid the imposition or fulfillment of a criminal penalty. It did not touch marriages at ages 16–17, which remain lawful with parental consent. Two features make this reform a revealing case. First, treatment is defined by age alone and applies nationwide from a precise date, so an age-eligibility contrast is available in the universe of registration records. Second—and this is the substantive point—the margin the law banned was already close to extinction. In the pre-reform years 2013–2018, about 3,400 fifteen-year-olds married in the entire country, a rate of roughly 4 per 100,000; in the same period, household survey data put the share of 15-year-old girls living in a coresident union between 1.2% (a conservative definition) and 4.8% (a broad one). These quantities are not directly comparable—a registration rate is a flow and union prevalence is a stock—but together they show that the reform targeted a small formal margin alongside a larger observed stock of coresident unions. Vital records add a third measurement margin: among mothers younger than 16 giving birth in 2019, 0.7% declared themselves formally married, while 19% declared a *união estável*, the Brazilian informal-union status.

This paper asks what a national ban accomplishes in that environment. We evaluate Law 13,811/2019 with a research design frozen before estimation: the specification, treatment window, comparison groups, and inference procedure were locked—with cryptographic hashes—before any post-reform coefficient was computed, and every subsequent amendment is logged. The design is a difference-in-differences by age eligibility on the universe of civil marriage registrations from 2013 to 2024: persons aged 15 (directly treated) are compared with persons aged 17–19 (untreated), within region and quarter, in a Poisson pseudo-maximum-likelihood model with population offsets, region-by-age and region-by-period fixed

effects, age-specific seasonality, and age-specific linear trends. Because the reform is a single national event, we do not treat regional cells as independent experiments: inference clusters on time periods and is triangulated with HAC-robust aggregate series, temporal bootstrap forecasts, placebo reform dates, and randomization inference, with all limitations reported.

Three results follow. First, the locked specification finds no detectable additional decline in marriage registrations at age 15 in the three fully treated quarters of 2019: the estimate is -1.1% , with a 95% confidence interval from -14.9% to $+14.9\%$, and an implied 2.1 registrations averted (CI $[-25.2, +33.8]$). This is an imprecise null, not evidence of a zero: the design’s minimum detectable effect at 80% power is a 19.3% decline. Conditional on the locked linear-trend specification, the confidence interval excludes declines larger than 14.9%. This is not a design-free bound: the same model without age-specific trends yields -37.8% ($p < 0.001$)—the age-15 series had been falling for years before the law, and any before–after comparison attributes that secular decline to the reform. We treat this divergence not as a robustness footnote but as a finding about how minimum-age laws can be mis-evaluated where the targeted margin is already declining.

Second, PNADC offers no evidence of a reduction in conservative coresident-union prevalence. In quarterly household survey data, the estimated prevalence among 15-year-olds does not fall after the reform; the point estimate is $+0.399$ percentage points (95% CI $[-0.003, +0.801]$, $p = 0.052$) relative to ages 17–19. The sign is consistent with substitution from formal marriage toward informal cohabitation, as documented for Mexico by Bellés-Obrero and Lombardi [2023], but we are explicit about what this coefficient can and cannot say: the registry measures a flow of formal events, the survey measures a stock of coresident arrangements, the two cannot be differenced, and equivalence testing shows the survey cannot rule out effects smaller than ± 0.5 p.p. The honest statement is that unions did not detectably decline, and the data lean toward relative informalization without establishing it.

Third, conjugal status at birth—observed for every live birth in the country, with the mother’s exact date of birth and municipality of residence—provides a separate view of the small formal margin. Among mothers under 16, the formally married share fell from 1.7% in 2013 to 0.9% in 2016 and 0.7% in 2019; the monthly event study shows this share collapsing through 2016 and then running flat at roughly -33% relative to its pre-2017 level, with no break at the law. Precisely because the decay is convex, the pre-registered difference-in-differences with group-specific *linear* trends returns $+29.3\%$ (95% CI $[+11.3\%, +50.1\%]$; $p = 0.0008$). The positive estimate runs against the reform’s direct eligibility mechanism, but the outcome is a stock conditional on giving birth, so composition can also move it. More decisively, the frozen diagnostics reject the design: the joint test of 24 leads has $p < 0.001$, and placebo reforms dated April 2017 and April 2018 “find” $+19.7\%$ and $+36.8\%$. The no-trend

variant returns -23.9% . We therefore report $[-23.9\%, +29.3\%]$ as a specification range, not a causal confidence interval. The fertility-composition contrast at age 15 is $+1.6\%$ (95% CI $[-1.3\%, +4.6\%]$; $p = 0.284$), and unknown-status reporting shows no detectable differential change ($p = 0.577$).

We make three contributions. To our knowledge, this is the first empirical evaluation centered on Brazil’s 2019 ban,¹ complementing the Mexican staggered-adoption evidence [Bellés-Obrero and Lombardi, 2023] with a single-national-reform design and a triangulation across formal-flow, coresident-stock, and status-at-birth measures. Substantively, we document a limiting case with general interest: a ban that arrives after the formal margin has nearly disappeared primarily constrains formalization and may leave union prevalence unchanged; its evaluation is dominated by pre-trends. This sharpens the interpretation of enforcement findings in Collin and Talbot [2023] and of null-to-perverse legal effects in Amirapu et al. [2026]. Methodologically, we show how much discipline a frozen specification buys in single-reform settings: the gap between -1.1% and -37.8% shows how inference changes when pre-existing age-specific trends are modeled.

The rest of the paper proceeds as follows. Section 2 reconstructs the legal change from primary sources. Section 3 describes the three data sources and the measurement decisions, which in this setting carry identification weight. Section 4 presents the design, the specification lock, and inference for a single national reform. Section 5 reports registry results; Section 6 turns to unions and to status at birth; Section 7 collects robustness and threats. Section 8 discusses what the results imply for minimum-age laws and their monitoring.

2 Institutional background

Under the Civil Code of 2002, the marriageable age in Brazil is 18, or 16–17 with the authorization of both parents or legal representatives (art. 1.517) [Brasil, 2002]. Before 2019, art. 1.520 allowed marriage *below* 16 in two exceptional situations: to avoid the imposition or fulfillment of a criminal penalty (a clause tied to the pre-2005 treatment of sexual offenses) and in case of pregnancy. Law 13,811, published and in force on March 13, 2019, rewrote art. 1.520 to eliminate both exceptions: since then, no civil marriage below 16 can be lawfully celebrated in Brazil [Brasil, 2019].

¹A systematic search on September 2, 2026—nineteen logged queries across economics (SSRN, NBER, IZA, CEPR, ANPEC), demography, public health (SciELO, PubMed), Brazilian legal scholarship, and forward citations of the closest international studies—found no causal evaluation of Law 13,811/2019. Existing empirical work is descriptive: Urquia et al. [2022] explicitly present their 2011–2018 perinatal series as a baseline against which the 2019 reform could later be measured, and a student report supervised by one of the authors (Faria, 2025, Insper) documents descriptively that the decline predates the law.

Three institutional facts matter for the design. First, the law changed eligibility for a formal act performed before a civil registrar; it created no new enforcement apparatus directed at informal unions, and *união estável* remains a constitutionally recognized family form with no minimum-age registration requirement. Second, the treated population is precisely delimited by age at the date of celebration: persons under 16. Ages 16–17 are legally untouched, though behaviorally they may absorb postponed marriages, which is why our primary comparison group is ages 17–19 and contrasts involving 16–17 are interpreted as potentially contaminated by delay. Third, the reform was national and immediate, with no phase-in, no state discretion, and no grandfathering of pending authorizations that we could verify in primary sources; the identification problem is therefore the single-event problem, not the staggered-adoption problem, and we treat it as such throughout.

The registry data also show what the law was regulating. Marriages with at least one spouse below 16 fell from 1,093 in 2013 to 512 in 2018—before any reform—and to 419 in 2019 and 193 in 2024. The decline that matters for Brazilian adolescents was well under way for reasons the law did not cause, which is precisely why an evaluation that does not model it will misattribute it.

3 Data and measurement

Our three sources measure different objects. We keep them separate by construction and never subtract a stock coefficient from a flow coefficient.

3.1 Civil marriage registrations (flow of formal events)

The Civil Registry statistics (IBGE, SIDRA table 4406) record every civil marriage by single year of age of each spouse, sex, month, and place of *registration*, from 2013 to 2024. We reconstructed the official aggregates exactly: all 3,780 local age–sex cells and every verifiable local total match the published figures. Age refers to age at registration in completed years; the month is the month of registration, not celebration; and the geography is the registry office, not the couple’s residence. Each of these measurement facts constrains the design: completed years rule out a conventional regression discontinuity, registration lags blur the treatment date (March 2019 is excluded; the post window opens in April/2019Q2), and office-versus-residence geography caps the defensible spatial resolution at the region level, per a precision rule fixed before estimation.

3.2 PNADC (stock of coresident unions)

The continuous national household survey (PNADC) provides quarterly cross-sections of 2.43 million adolescents aged 14–19 over 2012–2024 (48 quarters; calibrated weights; design-based variance using strata and PSUs). Our conservative union indicator equals one when the adolescent is the spouse/partner of the household head or is the head with a coresident partner; a broader indicator adds plausible nested pairings and is used only as robustness with an explicit ambiguity flag. The survey follows dwellings, not individuals; an adolescent who leaves home to form a union elsewhere may exit the sampled dwelling, so the indicator undercounts unions in a way that can correlate with the outcome. We therefore treat PNADC as measuring the stock of *coresident, household-grid-visible* unions, and we report equivalence tests and minimum detectable effects rather than interpreting non-significance as absence of effect.

3.3 SINASC (conjugal status at birth)

The national live-birth information system (SINASC, Ministry of Health) records every live birth with the mother’s age—and, since the 2011 form, her exact date of birth—her declared conjugal status (single; married; widowed; separated/divorced; *união estável*; unknown), municipality of *residence*, and the child’s date of birth, along with education, race, parity, and prenatal-care covariates. We use the official open microdata, 2013–2024 (2.4–2.9 million births per year; SHA-256-verified acquisition; layouts harmonized across the 2015/2016 file-generation break; audit in the replication package). Cross-validation against published SVSA totals confirmed the open files to the record for 2014 but revealed the 2015 export to be materially incomplete; 2015 is excluded from estimation under a logged data-error amendment (Section 6). Two caveats govern interpretation. Conjugal status is declared at the birth, without registry verification: it measures the stock of statuses among women giving birth, not a marriage flow, and post-law declarations could re-label rather than reflect behavior. And any SINASC estimand conditions on a birth having occurred, so it is a joint outcome of fertility and formalization; we present it as such. In exchange, SINASC observes 55–70 thousand births to mothers under 16 per year—against 194 treated-cell marriage registrations in 2019Q2–Q4—with residence geography and daily dates, which no other source offers.

3.4 Denominators and comparability

Population denominators by age, sex, region, and quarter come from PNADC with a cell-quality rule (minimum unweighted counts and maximum coefficients of variation) frozen before estimation; state-level cells failed the rule and region was selected as the primary geography. The under-15 registry category is reported as counts and shares only—no rate with a fabricated 0–14 population is ever computed.

4 Empirical strategy

4.1 Design and primary specification

Treatment is age eligibility: a 15-year-old could, in principle, marry under the pregnancy exception before March 13, 2019 and cannot since; a 17-year-old’s legal regime is unchanged. The primary estimand is the short-run change in civil marriage registrations of 15-year-olds per 100,000 residents aged 15, in 2019Q2–Q4, relative to ages 17–19, in ratio and in level terms.

The locked model is a PPML regression on $\text{region} \times \text{age} \times \text{quarter}$ counts (540 cells; 5 regions; 27 quarters; ages 15 and 17–19) with the log resident population as offset; region-by-age, region-by-period, and age-by-calendar-quarter fixed effects; age-specific linear trends; and the treatment indicator ($\text{age } 15 \times \text{post}$). 2019Q1 is omitted (partial treatment); 2020–2021 is flagged as pandemic-contaminated and the primary window ends in 2019Q4. Alternative control sets (18–19; 16–17), no-trend and linear-rate variants, monthly frequency, and pre-window starts in 2014 and 2015 were all frozen as mandatory robustness before any estimate existed.

4.2 Inference for one national reform

The reform is a single national event: regional cells share one treatment date and do not constitute independent experiments. Primary inference clusters on time periods (27 clusters, 3 post). We triangulate with (i) an aggregate Brazil series with HAC standard errors, (ii) a temporal block bootstrap of the pre-period forecast, (iii) placebo reform dates in the pre-period, and (iv) randomization inference over feasible placebo dates, whose resolution with only four dates ($p\text{-floor} = 1.00$) we report rather than hide. A two-way cluster variant produced an implausibly small standard error after a positive-semidefinite adjustment with five regions; it is reported as a diagnostic and given no evidentiary weight.

4.3 The specification lock

The full specification—estimands, samples, windows, comparison groups, model, and inference—was frozen and hash-stamped before any post-reform coefficient was computed (2026-09-02T00:56, SHA-256 in the replication package). Four amendments, all logged before the corresponding re-estimation, concern collinearity in saturated event studies and one exact computational substitution; none was motivated by a sign, magnitude, or significance. The SINASC estimands entered through the same mechanism: a separate extension lock (v1.0.0) was frozen and hash-stamped on 2026-09-02, after the data audit but before any post-reform SINASC contrast was computed, with expected signs recorded *ex ante*, a disclosure of the descriptive quantities already seen, and a mandatory diagnostic battery—whose results are reported in Section 6 exactly because they reject the design.

5 Registry results

5.1 Raw series and the vanishing margin

Figure 1 plots registration rates by single age. The age-15 series is one to two orders of magnitude below ages 17–19 throughout and declines over the whole pre-period. In the 2013–2018 window, 3,394 fifteen-year-olds married nationwide (4.19 per 100,000 per year), against 94,107 sixteen-year-olds and 130,000 seventeen-year-olds. The under-15 category shrinks from 364 marriages in 2013 to 246 in 2016 and 85 in 2024.

5.2 The locked estimate

The primary PPML contrast for 2019Q2–Q4 yields a rate ratio of 0.989: -1.1% , 95% CI $[-14.9\%, +14.9\%]$, $p = 0.888$ (period clusters). In levels this is -0.022 per 100,000, or 2.1 registrations averted against a counterfactual, with CI $[-25.2, +33.8]$. The aggregate HAC series gives $+2.0\%$ ($p = 0.785$) and the temporal bootstrap forecast $+2.0\%$ ($p = 0.857$). Figure 2 shows the event study: no post-reform break stands out from the pre-period variation.

5.3 Trend dependence as a result

Removing the age-specific linear trends flips the estimate to -37.8% ($p < 0.001$). Monthly frequency with trends gives -2.4% ($p = 0.820$). Starting the pre-window in 2014 or 2015 moves the locked estimate to $+8.2\%$ and $+11.1\%$. A synthetic age control built on pre-2019 data places all weight on age 17 and shows a short-run gap of -0.658 per 100,000, which

Taxas brutas de registros civis por idade
Brasil, ambos os sexos; escalas verticais livres por idade

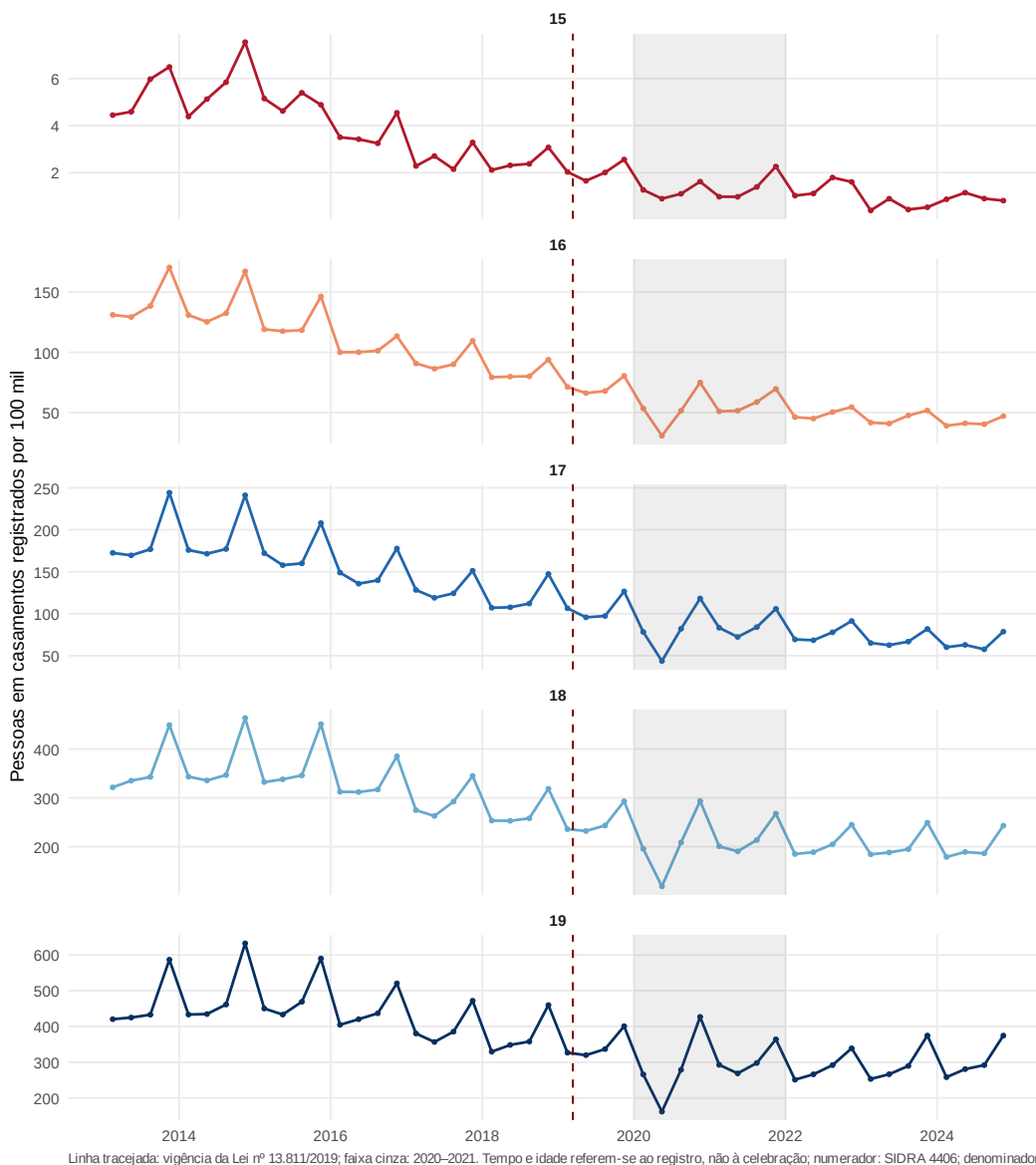


Figure 1: Civil marriage registrations per 100,000 residents, by single age at registration. Vertical line: Law 13,811 (March 2019). Source: SIDRA 4406 and PNADC denominators; replication package.

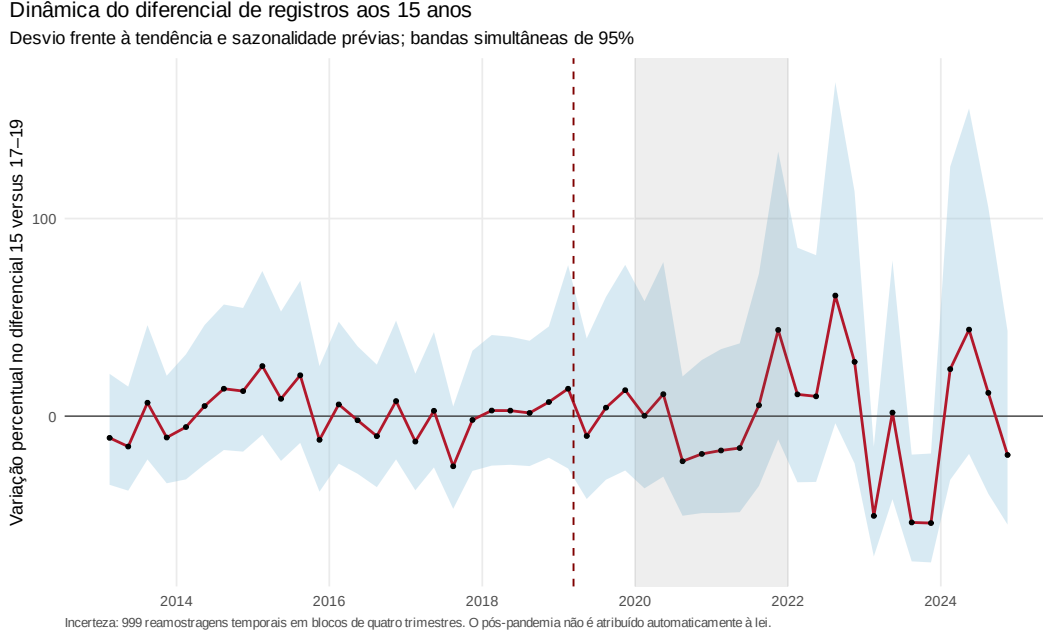


Figure 2: Event study, age 15 versus 17–19, locked specification. 2019Q1 omitted; post window 2019Q2–Q4. Source: replication package.

we read as descriptive corroboration, not new identification. The joint test of 24 leads does not reject ($p = 0.161$), but no simultaneous band stays within $\pm 10\%$, and one placebo date (2015Q2) is significant ($p = 0.002$): parallel evolution cannot be certified, it can only be modeled, and the conclusion depends materially on how. This is the central empirical lesson of the paper: in a collapsing series, the counterfactual trend *is* the estimate.

5.4 Power, and the absence of recapture

The design’s 80%-power MDE is a 19.3% decline (power is 28% against a 10% decline and 83% against 20%). Effects at ages 16–19 are +0.6%, +3.3%, +0.8%, and -2.4% , none surviving familywise correction; the aggregate “recapture” ratio (excess at 16–17 over deficit at 15) is undefined in half of the bootstrap draws and spans $[-114, +1,107]$ where defined. There is no reliable evidence of postponement toward legal ages, and with 2.1 point-estimated averted events there is little to recapture. By sex, the female estimate is -9.7% (Holm $p = 0.125$); the male +155.8% is driven by 36 zero cells on a rare base and is not interpretable.

Table 1: Locked primary age-based PPML estimate, 2019Q2–Q4.

rate_ratio	rate_ratio_ci_lower	rate_ratio_ci_upper	percent_change	percent_change_ci_lower	percent_change
0.989	0.851	1.149	-1.072	-14.853	

6 Unions and formalization at birth

6.1 Coresident unions (PNADC)

The point estimate for the conservative union indicator among 15-year-olds is positive after the reform. The design-based cell estimate is +0.399 p.p. (95% CI $[-0.003, +0.801]$, $p = 0.052$); the aggregate HAC series gives +0.235 p.p.; microdata LPMs with household-period clustering or full survey-design variance give +0.344 p.p. (CIs $[-0.119, +0.807]$ and $[-0.025, +0.712]$); the broad indicator gives +0.597 p.p. ($p = 0.102$). The MDE is 0.575 p.p. and equivalence at ± 0.50 p.p. fails (TOST $p = 0.311$). Estimates shrink as the pre-window shortens (0.399, 0.329, 0.113 p.p. for windows starting 2013, 2014, 2015). We conclude: no evidence of a reduction in conservative coresident-union prevalence; a positive, window-sensitive signal consistent with relative informalization that the survey cannot establish at conventional strength.

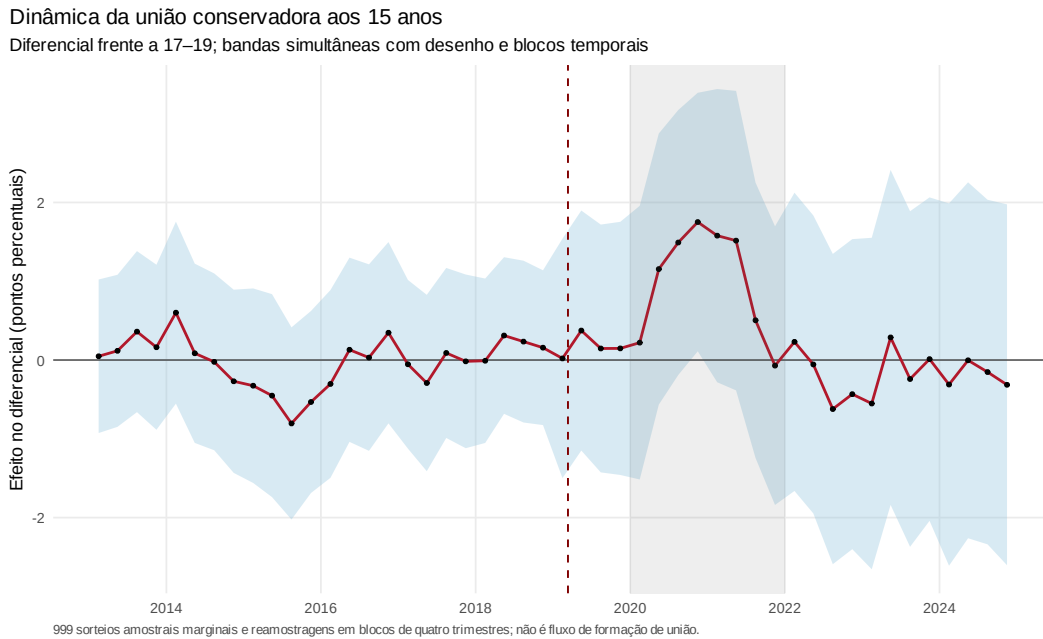


Figure 3: Coresident-union prevalence at age 15 (conservative definition), event study relative to ages 17–19. Source: replication package.

6.2 Conjugal status at birth (SINASC)

Among the roughly 55–80 thousand mothers under 16 giving birth each year, the formally married share was already small and falling before the law: 1.72% in 2013 (1,345 of 78,062), 0.88% in 2016 (593 of 67,419), and 0.69% in 2019 (374 of 54,166), against informal-union

shares of 27.9%, 23.2%, and 19.5%. Mothers aged 16–17 in 2019: 3.9% married, 23.3% in *união estável*. Monthly counts of married under-16 mothers in 2019 (19–47 per month) show no visible April collapse, consistent with status-at-birth being a slowly decaying stock among parturients rather than a marriage flow. Validation against the official SVSA epidemiological bulletin uncovered a material defect in the open microdata: the 2014 file matches the published total to the record (2,979,259), but the 2015 export is short by an implied 231,143 records (7.7%), concentrated in October–December; calendar year 2015 is therefore excluded from every estimate below, under a data-error amendment logged before re-estimation.

The pre-registered contrast (extension lock v1.0.0: mothers 10–15 versus 17–19, region-of-residence by month PPML on counts of married births with valid-status births as exposure, the registry’s fixed-effect and trend structure, period clustering) returns +29.3% (95% CI [+11.3%, +50.1%], $p < 0.001$; 270 married births in the treated post cells; Appendix Tables 2–3). We recorded the expected sign as negative-or-null before estimation. The positive estimate runs against the direct eligibility mechanism, while conditioning on birth leaves room for composition; the frozen diagnostics decide whether the contrast is credible, and they reject it. The monthly event study (Figure 4) shows the treated married share falling steeply through 2016 and then running flat at -30 to -37% of its pre-2017 level, straight across the reform date; the joint test of 24 leads has $p \approx 0$; and placebo bans dated April 2016, 2017, and 2018 return -5.9% (n.s.), $+19.7\%$ ($p = 0.004$), and $+36.8\%$ ($p < 0.001$)—a monotone-in-time pattern that is the signature of a linear trend extrapolating a convex, floor-approaching decay, not of treatment. The no-trend variant returns -23.9% ; every trended variant (alternative controls, treated age 15 only, exact maternal age from dates of birth, quarterly aggregation, missing-as-unmarried) sits between $+22\%$ and $+33\%$ —except the variant whose pre-window starts in 2016, after the flattening, which returns $+1.5\%$ ($p = 0.86$), exactly as the artifact reading predicts. We therefore report $[-23.9\%, +29.3\%]$ as a specification range, not a causal confidence interval. The *união-estável* and any-union contrasts ($+3.7\%$ and $+5.1\%$) inherit the same defect and receive no causal reading. Two frozen diagnostics survive: the fertility-composition contrast—births to mothers aged 15 per 100,000 girls aged 15—is $+1.6\%$ (95% CI $[-1.3\%, +4.6\%]$; $p = 0.284$), and unknown-status reporting shows no detectable differential change ($p = 0.577$). What SINASC establishes, then, is descriptive and disciplinary at once: the formal margin at birth was small and had stopped falling *before* the ban, and even a universe-sized administrative dataset cannot rescue a short-run contrast whose counterfactual trend is the conclusion.

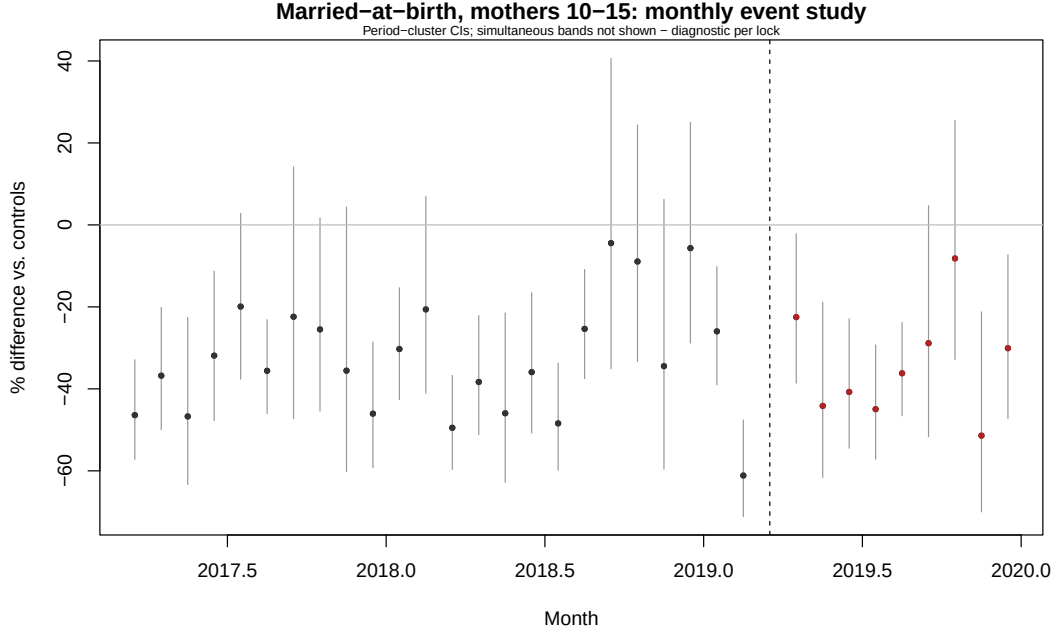


Figure 4: Married-at-birth among mothers 10–15 versus 17–19: monthly event study relative to the pooled pre-2017 baseline. Period-cluster confidence intervals; diagnostic per the extension lock. Source: replication package.

7 Robustness and threats to identification

The threats we can address are addressed by design: registration lag (omit March/Q1; monthly variant), pandemic contamination (primary window ends 2019Q4; 2020–2021 flagged), geographic mismatch (region-level analysis; municipal contrasts blocked), denominator uncertainty (repeated draws; SD an order below the temporal SE), and multiplicity (Holm within families). A pre-exposure triple-difference—regions with higher pre-law shares of under-16 marriages—shows a positive but imprecise gradient (+9.9% per SD, $p = 0.332$) and a -0.618 holdout correlation between exposure and 2018 changes, indicating mean reversion; it does not identify the national effect and is reported as descriptive. The threats we cannot address are named rather than assumed away: anticipation between the bill’s passage and enforcement; age measured in completed years; nationally synchronized age-specific shocks, which no fixed effect absorbs; and the selection inherent in any status-at-birth outcome.

8 Discussion and conclusion

Brazil banned civil marriage below 16 after registrations at that age had become rare, while coresident unions remained more prevalent on a conceptually different stock measure. Our frozen design finds no detectable additional decline in the banned formal events

and no detectable reduction in conservative coresident-union prevalence, with a suggestive—unproven—tilt toward informality, while the status-at-birth margin had flattened before the reform and its pre-registered contrast fails the placebo battery, leaving a specification range that spans zero.

Two implications travel beyond Brazil. For policy, minimum-age laws are instruments aimed at the recorded margin; where unions are informal, the recorded margin can go to zero while the phenomenon continues, and the case for complementary instruments (education, income support, enforcement against coerced unions) is correspondingly stronger. For evaluation and monitoring, the gap between our locked -1.1% and the naive -37.8% is a warning: registered child marriages were falling secularly in many countries before their bans, and legal-impact studies that do not model that trajectory can overstate what the law achieved. The status-at-birth exercise shows the mirror image—a mis-specified trend can just as easily manufacture a *positive* “effect” of a ban, and only a pre-committed placebo battery tells the two apart—a bias that survey-based monitoring of the SDG target, which measures unions rather than marriages, will not correct in the opposite direction. Honest evaluation in this literature requires pre-committed specifications, explicit power statements, and the discipline of keeping flows, stocks, and statuses apart.

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A SINASC status-at-birth exhibits

Table 2: SINASC status-at-birth contrasts, mothers 10–15 versus 17–19, 2019M4–M12 (extension lock v1.0.0). The frozen diagnostics reject a causal reading of these contrasts (Table 3); they are reported as specification-dependence evidence.

Outcome	% effect	95% CI	p (Holm)	No-trend %	MDE ₈₀ %	TOST p	Treated ever
Married (S1)	29.3	[11.3; 50.1]	0.002	-23.9	23.8	0.892	270
União estável (S2)	3.7	[0.9; 6.7]	0.010	-3.7	4.1	<0.001	7,848
Any declared union (S3)	5.1	[2.2; 8.0]	0.001	-3.7	4.0	<0.001	8,118

PPML on region-of-residence $\times$ group $\times$ month cells; offset = births with valid status; period-cluster inference; group-specific linear trends (column 2) or none (column 5). TOST margin: rate ratio in [0.85; 1.176].

Table 3: SINASC frozen diagnostics: placebo reform dates, lead test, aggregate HAC, misreporting, and fertility-composition contrasts.

Object	% effect	95% CI	p
Placebo ban April 2016	-5.9	[-21.1; 12.2]	0.497
Placebo ban April 2017	19.7	[5.9; 35.3]	0.004
Placebo ban April 2018	36.8	[16.6; 60.6]	<0.001
Brazil monthly gap, HAC(12)	24.8	[3.5; 50.4]	0.024
Unknown-status share (misreporting)	3.4	[-8.1; 16.3]	0.577
Fertility, age 15 (S4)	1.6	[-1.3; 4.6]	0.284
Fertility, age 14 (S4 rob.)	-3.0	[-6.1; 0.2]	0.137
Joint test, 24 monthly leads		$p < 0.001$	

Placebo bans use pre-2019 data only, with the true model and windows transposed to each placebo year.

Fertility rows: PPML per 100,000 female residents of the same age (PNADC denominators), quarterly, Holm within family.

B Registry and PNADC exhibits

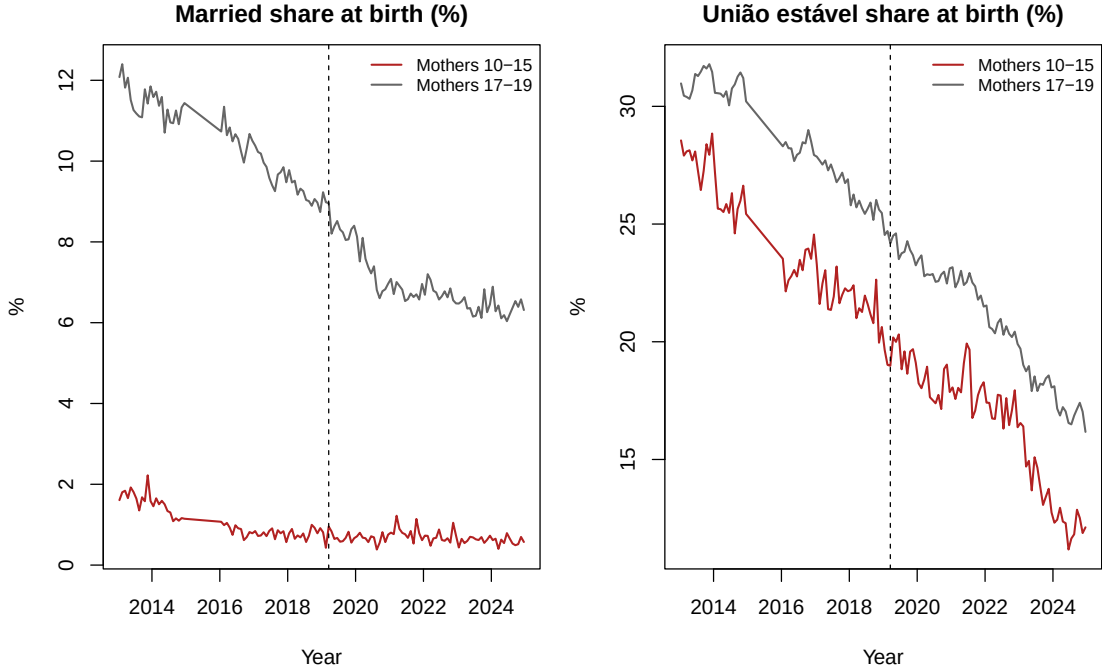


Figure 5: Declared conjugal status at birth, Brazil, monthly 2013–2024: married and *união estável* shares among mothers aged 10–15 and 17–19. Vertical line: Law 13,811 (March 2019). Source: replication package.

Table 4: Trend and control-group robustness. Period-cluster standard errors.

	variant	controls	estimate	std_error	p_value	rate_ratio	percent_change
Primary: age trends		17-19	-0.011	0.077	0.888	0.989	-1.072
No age trends		17-19	-0.476	0.073	0.000	0.622	-37.847
Alternative controls with age trends		18-19	-0.003	0.079	0.971	0.997	-0.290
Alternative controls with age trends		16-17	-0.033	0.070	0.637	0.967	-3.266

Table 5: Age-specific event effects and aggregate recapture. The ratio is unstable and does not track people.

age	controls	percent_change	estimated_event_effect	event_effect_ci_lower	event_effect_ci_upper	aggregat
15	17-18-19	-1.072	-2.102	-48.318	31.834	
16	18-19	0.609	40.567	-457.688	518.075	
17	18-19	3.321	326.852	-135.267	781.018	

Table 6: Primary design-based cell estimate for conservative co-resident union.

effect_percentage_points	combined_se_quadrature	ci_lower_percentage_points	ci_upper_percentage_points	p
0.399	0.002	-0.003	0.801	

Table 7: Inference triangulation for the national reform. Estimand scales differ where labeled.

	method	estimand_scale	est
	PPML cluster by period (primary)	log rate ratio	-
	PPML two-way region and period	log rate ratio	-
	PPML cluster by region (diagnostic)	log rate ratio	-
	Brazil age15-vs-pooled17-19 log-rate gap; Newey-West lag 4	aggregated log-rate gap	-
pre-period linear-seasonal forecast with moving-block residual bootstrap	aggregated forecast log-rate gap		-
pre-specified date randomization inference	studentized PPML statistic		-

Table 8: Minimum detectable effects and power diagnostics.

outcome	target_decline_percent	target_log_rate_ratio	approximate_power	standard_error
formal marriage age 15	10	-0.105	0.280	0.077
formal marriage age 15	20	-0.223	0.830	0.077
formal marriage age 15	30	-0.357	0.997	0.077
formal marriage age 15	40	-0.511	1.000	0.077
union_conservative age 15			0.800	0.002